QUICK START & SAFETY GUIDE · XR · XR-16

From the Box to Live Signals in VR

Seven steps: fit the mask, mount the BCI box, power it from a battery, clean the skin, connect from the headset, a browser or a phone, and check the signal with a blink test.



WARNING Battery only.

Power the BCI box from a **5 V power bank or a battery head strap**. Never from a wall charger or a computer on mains power, and never while the headset is charging. PiEEG XR is **not a medical device**.

01 SAFETY RULES — READ BEFORE PUTTING THE MASK ON ANYONE

#	Rule
1	**Battery only.** Power bank or battery head strap; never a wall charger or a cable to a computer on mains power.
2	**No diagnosis.** Attention, relaxation and expression values are research indicators, not a measure of any condition.
3	**Implanted devices.** Not on a person with a pacemaker or other implant without a physician's approval.
4	**Skin.** Clean, intact skin only. Stop if it becomes irritated; clean off any gel after use.
5	**VR safety.** Follow the headset maker's safety guidance, clear the play area and take breaks.
6	**Flashing.** Update firmware with the mask off the face.
7	**Hygiene.** Wipe electrodes and silicone with an alcohol pad before and after each user.

Power while worn

Power source	While worn	Why
Power bank or battery head strap	SAFE	No path to the mains.
Headset's own battery, headset not charging	SAFE	Still fully on battery.
Wall charger	NEVER	Connects the person to the mains through the electrodes.
USB cable to a PC on mains power	NEVER	Same reason.
Headset plugged in to charge	NEVER	The charger links the mains to the headset and the mask.

For research and education only. Not a medical device. Specifications subject to change.

02 WHAT YOU NEED

Item	Requirement
PiEEG XR	Mask and BCI box made for your headset model
5 V source	Charged power bank or battery head strap with a USB output
Cable	USB-A to USB-C
Alcohol pads	For the skin and the electrodes; gel or saline optional
Receiver	BodyPress on the Quest or an Android phone, or Chrome / Edge on a computer

03 SETUP IN SEVEN STEPS

1 Replace the face mask

Remove the original facial interface, align the four buckles, snap in the two back and then the two front buckles (XR-MD-002).

2 Mount the BCI box

Detach the head strap, hold the box against the headset, reattach the strap so it clamps the box, route the mask cable.

3 Power on

Connect the power bank or battery strap with the USB-A to USB-C cable. The box advertises as **PiEEG XR**.

4 Clean and put it on

Wipe the forehead, temples and electrodes with alcohol, let them dry, put the headset on and tighten the strap until the mask presses evenly.

5 Connect

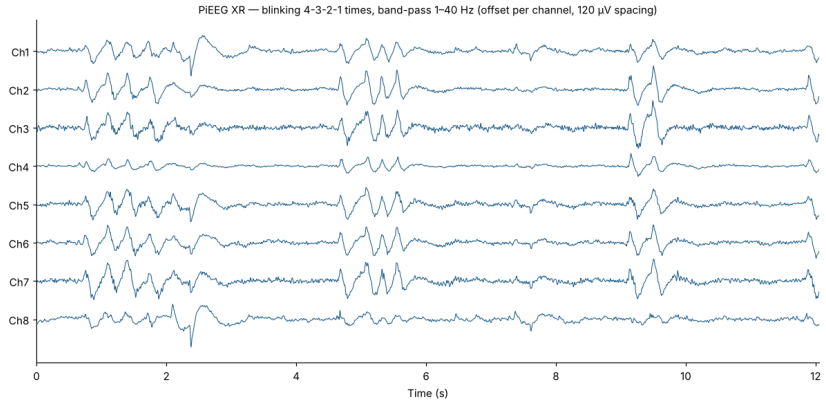
Quest: BodyPress from the Meta Quest Store → connect to PiEEG XR. **Browser:** cloud.pieeg.com in Chrome or Edge → PiEEG Devices → PiEEG XR (or XR-16). **Android:** BodyPress from Google Play.

6 Let it settle

Sit still and relax the face for a few seconds while the filter settles.

7 Blink test

Blink hard 3–5 times: large, sharp spikes on the forehead channels. Look left and right: opposite swings on the side channels. Clench the jaw: a burst of fuzz.



Blink test: groups of 4, 3, 2 and 1 blinks on all 8 channels, band-pass 1–40 Hz.

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04 TROUBLESHOOTING

Power and Bluetooth

Problem	Fix
Device not visible	Usually power: use a USB-A to USB-C cable. Open the lid and check that an LED lights inside; no LED — try another cable or power bank.
Visible but no data	Update the firmware (XR-SW-006, section 07).
No Bluetooth in the browser	Use Chrome or Edge; Safari, Firefox and iPhone / iPad browsers lack Web Bluetooth.
PiEEG Server does not find it	pip install "pieeg-server[ironbci]", then pieeg-server --device ironbci8 --ble-name "PiEEG XR".
Connection drops	Move the receiver closer and close other Bluetooth apps.

Signal and mask

Problem	Fix
A channel is a flat line	That electrode is not touching. Reseat the mask, tighten the strap, add a little gel.
All channels railed or identical	The reference or bias (centre forehead) has lost contact. Reseat the mask.
Thick fuzzy band on every channel	50/60 Hz hum: battery only, move away from chargers, use the notch filter.
Blinks do not show	The forehead electrodes are not touching. Clean the skin and fit the mask firmly.
Mask shifted or insert loose	Push the silicone back so it sits evenly all the way round.

05 RELATED DOCUMENTS

XR-DS-001
Product Datasheet

XR-MD-002
Mechanical & Assembly Drawing

XR-PCB-003
PCB & Manufacturing Specification

XR-PIN-004
Mask, Electrode & Connector Guide

XR-SW-006
Bluetooth Interface & Software Guide

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